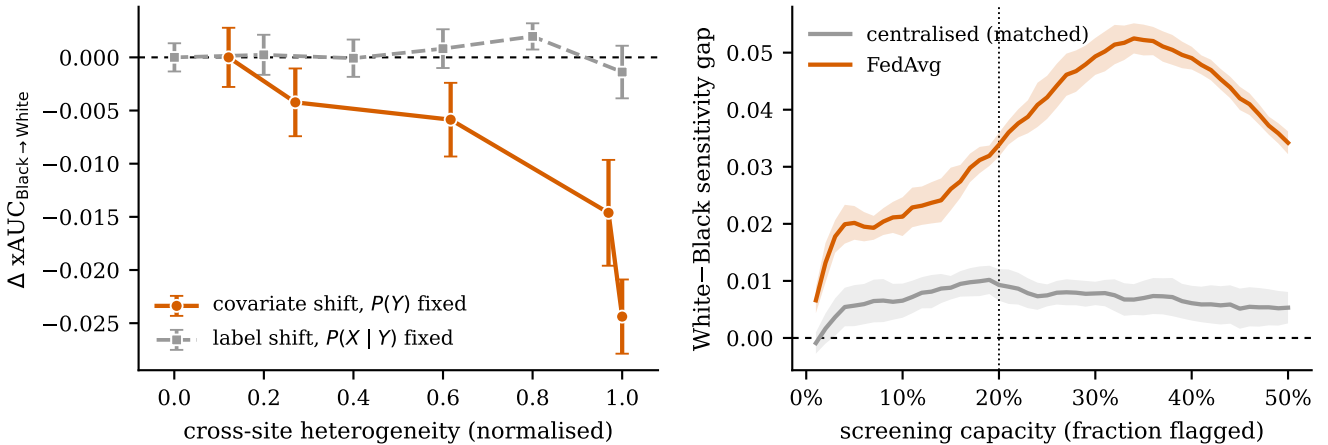
(a) Which heterogeneity moves ranking
Δ xAUC$_{\text{Black}\to\text{White}}$
0.000
−0.005
−0.010
−0.015
−0.020
−0.025
covariate shift, $P(Y)$ fixed
label shift, $P(X \mid Y)$ fixed
0.0
0.2
0.4
0.6
0.8
1.0
cross-site heterogeneity (normalised)
(b) Not an artefact of one threshold
centralised (matched)
FedAvg
White−Black sensitivity gap
0.05
0.04
0.03
0.02
0.01
0.00
0%
10%
20%
30%
40%
50%
screening capacity (fraction flagged)